Econometrics Cheat Sheet

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The Econometrics Cheat Sheet Project

Basic concepts

Definitions

Econometrics: Social Sciences discipline with the objective of quantifying the relationships between economic agents, testing economic theories and evaluating and implementing government and business policies.
Econometric model: Simplified representation of reality to explain economic phenomena.
Ceteris paribus: If all the other relevant factors remain constant.

Data structures

Cross-section: Sample taken at a given point in time, a static *photo*. Order does not matter.
Time series: Observations over time. Order does matter.
Panel data: A time series for each observation of a cross-section.
Pooled cross-sections: Cross-sections from different time periods.

Phases of an econometric model

1. Specification.
2. Estimation.
3. Validation.
4. Utilization.

Regression analysis

Study and predict the mean value of a variable (dependent, y) based on fixed values of other variables (independents, x's). It is very common to use Ordinary Least Squares (OLS).

Correlation analysis

Correlation analysis does not distinguish between dependent and independent variables.

- Simple correlation measures the degree of linear association between two variables.
$$r = \frac{\mathrm{Cov}(x,y)}{s_x \cdot s_y} = \frac{\sum_{i=1}^n ((x_i-\overline{x})\cdot(y_i-\overline{y}))}{\sqrt{\sum_{i=1}^n (x_i-\overline{x})^2 \cdot \sum_{i=1}^n (y_i-\overline{y})^2}}$$
- Partial correlation measures the degree of linear association between two variables controlling for a third.

Assumptions and properties

OLS assumptions

Gauss-Markov assumptions:

1. **Linearity in parameters**. y must be a linear function of the β's.
2. **Random sampling**. The sample has been randomly taken (only when cross-section).
3. **No perfect collinearity**:
   - There are no independent variables that are constant: $\mathrm{Var}(x_j) \neq 0 \quad \forall j = 1, \ldots, k$
   - There is no exact linear relation between independent variables.
4. **Conditional mean zero and correlation zero**:
   a. There are no relevant variables left out of the model: $\mathrm{Cov}(x_j, u) = 0 \quad \forall j = 1, \ldots, k \rightarrow$ **exogeneity**
   b. There are no systematic errors: $\mathrm{E}(u \mid x_1, \ldots, x_k) = \mathrm{E}(u) = 0$ $\rightarrow$ **strict exogeneity** (4a holds)
5. **Homoscedasticity**: The variability of the error term is the same for all levels of x:
$$\mathrm{Var}(u_i \mid x_{1i}, \ldots, x_{ki}) = \sigma^2 \quad \forall i = 1, \ldots, n$$
6. **No autocorrelation**: Errors do not contain information about any other errors:
$$\mathrm{Corr}(u_t, u_s \mid x_1, \ldots, x_k) = 0 \quad \forall t \neq s$$
7. **Normality**: Errors are independent and identically distributed: $u \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$
8. **Data size**: The number of observations available must be greater than $(k+1)$ parameters to estimate.

Asymptotic properties of OLS

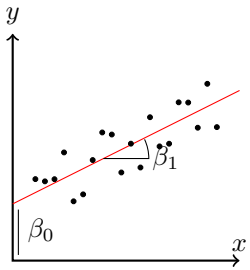
Under the Gauss-Markov assumptions and the Central Limit Theorem (CLT):

- Hold 1 to 4a: OLS is **consistent**: $\mathrm{plim}(\hat{\beta}_j) = \beta_j$
- Hold 1 to 4b: OLS is **unbiased**: $\mathrm{E}(\hat{\beta}_j) = \beta_j$
- Hold 1 to 5: **Asymptotic normality** of estimators: $\hat{\beta}_j \underset{a}{\sim} \mathcal{N}(\beta_j, \mathrm{Var}(\hat{\beta}_j))$
- Hold 1 to 6: **Unbiased estimate** of σ^2: $\mathrm{E}(\hat{\sigma}^2) = \sigma^2$
- Hold 1 to 6: OLS is BLUE (Best Linear Unbiased Estimator) or **efficient**.
- Hold 1 to 7: Hypothesis testing and confidence intervals can be done reliably.

Ordinary Least Squares

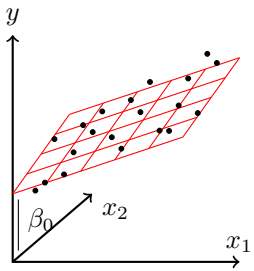
Objective: Minimise the Sum of Squared Residuals: $\min \sum_{i=1}^n \hat{u}_i^2$, where $\hat{u}_i = y_i - \hat{y}_i$

Simple regression model



Equation:
$$y_i = \beta_0 + \beta_1 x_i + u_i$$
Estimation:
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$
where:
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}$$
$$\hat{\beta}_1 = \frac{\mathrm{Cov}(y,x)}{\mathrm{Var}(x)}$$

Multiple regression model



Equation:
$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$
Estimation:
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \cdots + \hat{\beta}_k x_{ki}$$
where:
$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}_1 - \cdots - \hat{\beta}_k \overline{x}_k$$
$$\hat{\beta}_j = \frac{\mathrm{Cov}(y, \tilde{x}_j)}{\mathrm{Var}(\tilde{x}_j)}$$
Matrix: $\hat{\beta} = (X^\top X)^{-1}(X^\top y)$

Interpretation of coefficients

Model	Dependent	Independent	β_1 interpretation
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y \approx (\beta_1/100)(\%\Delta x)$
Log-level	$\log(y)$	x	$\%\Delta y \approx (100\beta_1)\Delta x$
Log-log	$\log(y)$	$\log(x)$	$\%\Delta y \approx \beta_1(\%\Delta x)$
Quadratic	y	$x + x^2$	$\Delta y \approx (\beta_1 + 2\beta_2 x)\Delta x$

Error measurements

Sum of Sq. Residuals: $\mathrm{SSR} = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
Explained Sum of Squares: $\mathrm{SSE} = \sum_{i=1}^n (\hat{y}_i - \overline{y})^2$
Total Sum of Sq.: $\mathrm{SST} = \mathrm{SSE} + \mathrm{SSR} = \sum_{i=1}^n (y_i - \overline{y})^2$
Standard Error of the Regression: $\hat{\sigma} = \sqrt{\frac{\mathrm{SSR}}{n-k-1}}$
Standard Error of the $\hat{\beta}$'s: $\mathrm{se}(\hat{\beta}) = \sqrt{\mathrm{diag}(\hat{\sigma}^2 \cdot (X^\top X)^{-1})}$
Root Mean Squared Error: $\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$
Mean Absolute Error: $\mathrm{MAE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$
Mean Abs. Percentage Error: $\mathrm{MAPE} = \frac{\sum_{i=1}^n |\hat{u}_i / y_i|}{n} \cdot 100$

R-squared

Measures the **goodness of the fit**, how the regression fits the data:

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

- Measures the **percentage of variation** of y that is linearly **explained** by the variations of x's.
- Takes values **between 0** (no linear explanation) **and 1** (total explanation).

When the number of regressors increases, the value of the R-squared also increases, whether the new variables are relevant or not. To solve this problem, there is an **adjusted R-squared** by degrees of freedom (or corrected):

$$\overline{R}^2 = 1 - \frac{n-1}{n-k-1} \cdot \frac{\text{SSR}}{\text{SST}} = 1 - \frac{n-1}{n-k-1} \cdot (1 - R^2)$$

For big sample sizes: $\overline{R}^2 \approx R^2$

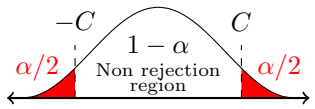
Hypothesis testing

Definitions

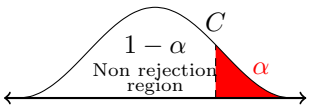
Rule designed to determine from a sample **whether there is evidence to reject a hypothesis** that is made about one or more population parameters. Elements:

- **Null hypothesis** (H_0): Hypothesis to be tested.
- **Alternative hypothesis** (H_1): Hypothesis that cannot be rejected when H_0 is rejected.
- **Test statistic**: Random variable whose probability distribution is known under H_0.
- **Critical value** (C): Value against which the test statistic is compared to determine if H_0 is rejected or not. It sets the frontier between the regions of non-rejection and rejection of H_0.
- **Significance level** (α): Probability of rejecting H_0 being true (Type I Error). Chosen by those who conduct the test. Commonly 10%, 5% or 1%.
- **p-value**: Lowest level of significance at which H_0 can be rejected.

Two-tailed test, H_0 distrtibution



One-tailed test, H_0 distribution



If p-value $< \alpha$ holds, there is evidence to reject H_0, thus, there is evidence to support H_1.

Individual tests

Tests whether a parameter is significantly different from a given value, ϑ.

- $H_0 : \beta_j = \vartheta$
- $H_1 : \beta_j \neq \vartheta$

$$\text{Under } H_0: \quad t = \frac{\hat{\beta}_j - \vartheta}{\text{se}(\hat{\beta}_j)} \sim t_{n-k-1}$$

If $|t| > |t_{n-k-1,\alpha/2}|$, there is evidence to reject H_0.

Individual significance test: Tests whether a parameter is significantly **different from zero**.

- $H_0 : \beta_j = 0$
- $H_1 : \beta_j \neq 0$

$$\text{Under } H_0: \quad t = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \sim t_{n-k-1}$$

If $|t| > |t_{n-k-1,\alpha/2}|$, there is evidence to reject H_0.

The F test

Simultaneously tests multiple (linear) hypotheses about the parameters. It makes use of a non-restricted model and a restricted model:

- **Non-restricted model**: Model on which we want to test the hypotheses.
- **Restricted model**: Model on which the hypotheses have been imposed.

Then, looking at the errors, there are:

- $\textbf{SSR}_{\textbf{UR}}$: SSR of the non-restricted model.
- $\textbf{SSR}_{\textbf{R}}$: SSR of the restricted model.

$$\text{Under } H_0: \quad F = \frac{\text{SSR}_\text{R} - \text{SSR}_\text{UR}}{\text{SSR}_\text{UR}} \cdot \frac{n-k-1}{q} \sim F_{q,n-k-1}$$

where k is the number of parameters of the non-restricted model (excluding the constant) and q is the number of linear hypotheses tested.

If $F > F_{q,n-k-1}$, there is evidence to reject H_0.

Global significance test: Tests whether all parameters associated with the x's are **simultaneously equal to zero**.

- H_0: $\beta_1 = \beta_2 = \cdots = \beta_k = 0$
- H_1: At least one $\beta_j \neq 0 \quad j = 1, \ldots, k$

Simplifying the formula for the F statistic:

$$\text{Under } H_0: \quad F = \frac{R^2}{1-R^2} \cdot \frac{n-k-1}{k} \sim F_{k,n-k-1}$$

If $F > F_{k,n-k-1}$, there is evidence to reject H_0.

Confidence intervals

The confidence interval at $(1-\alpha)$ confidence level:

$$\hat{\beta}_j \pm |t_{n-k-1,\alpha/2}| \cdot \text{se}(\hat{\beta}_j)$$

Dummy variables

Dummy (or binary) variables are used for qualitative information like sex, civil status, country, etc.

- Take the **value 1** in a given category and **0 in the rest**.
- They are used to analyse and model **structural changes** in the parameters.

If a qualitative variable has m categories, only $(m-1)$ dummy variables must be included in a model with an intercept (include all m if the model has no intercept).

Structural change

Structural change refers to changes in the values of the parameters of the econometric model produced by the effect of different sub-populations. Structural change can be included in the model through dummy variables.

The location of the dummy variables (D) matters:

- **On the intercept** (additive effect): Represents the mean difference between the values produced by the structural change.

$$y = \beta_0 + \delta_1 D + \beta_1 x_1 + u$$

- **On the slope** (multiplicative effect): Represents the effect (slope) difference between the values produced by the structural change.

$$y = \beta_0 + \beta_1 x_1 + \delta_1 D \cdot x_1 + u$$

Chow's structural test: Tests the existence of structural changes in all the model parameters. It is a particular expression of the F test, where H_0: No structural change (all $\delta = 0$).

Changes of scale

Changes in the **measurement units** of the variables:

- **Endogenous**: $y^* = y \cdot \lambda \rightarrow \beta_j^* = \beta_j \cdot \lambda \quad \forall j = 0, \ldots, k$
- **Exogenous**: $x_j^* = x_j \cdot \lambda \rightarrow \beta_j^* = \beta_j / \lambda$
- Endogenous and exogenous at the same time: $y^* = y \cdot \lambda$ and $x_j^* = x_j \cdot \lambda \rightarrow \beta_0^* = \beta_0 \cdot \lambda$; $\beta_j^* = \beta_j$.

Changes of origin

Changes in the **measurement origin** of the variables only affects the model's intercept:

- **Endogenous**: $y^* = y + \lambda \rightarrow \beta_0^* = \beta_0 + \lambda$
- **Exogenous**: $x_j^* = x_j + \lambda \rightarrow \beta_0^* = \beta_0 - \lambda\beta_j$

Multicollinearity

- **Perfect multicollinearity**: There are independent variables that are constant and/or there is an exact linear relation between independent variables. This **breaks the third (3)** OLS **assumption**.
- **Approximate multicollinearity**: There are independent variables that are approximately constant and/or there is an approximately linear relation between independent variables. This **does not break any** OLS **assumption**, but there are effects.

Consequences

- **Perfect multicollinearity**: The equation system of OLS cannot be solved due to infinite solutions.
- **Approximate multicollinearity**
  - Small sample variations can induce big variations in the OLS estimations.
  - The variance of the OLS estimators of the x's that are collinear, increases, thus the inference of the parameter is affected. The estimation of the parameter is very imprecise (big confidence interval).

Detection

- **Correlation analysis**: Look for high correlations between independent variables, $|r| > 0.7$.
- **Variance Inflation Factor (VIF)**: Indicates the increase of $\text{Var}(\hat{\beta}_j)$ because of the multicollinearity.

$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1-R_j^2}$$

  where R_j^2 denotes the R-squared from a regression between x_j and all the other x's.
  - Values between 4 to 10: Warning.
  - Values > 10: Serious problem.

One typical characteristic of multicollinearity is that the regression coefficients of the model are not individually different from zero (due to high variances), but jointly they are different from zero.

Correction

- Delete one of the collinear variables.
- Perform factor analysis (or any other dimension reduction technique) on the collinear variables.
- Interpret coefficients with multicollinearity jointly.

Heteroscedasticity

The error term u_i does not have a constant variance conditional to the explanatory variables:

$$\text{Var}(u_i \mid x_{1i}, \ldots, x_{ki}) = \sigma_i^2 \neq \sigma^2$$

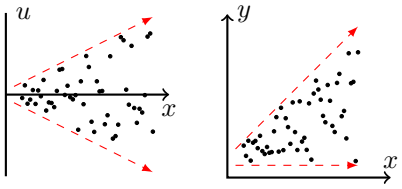
This **breaks the fifth (5)** OLS **assumption**.

Consequences

- OLS estimators are still consistent.
- OLS estimators are still unbiased.
- OLS is **no longer efficient**, but still a LUE (Linear Unbiased Estimator).
- **Variance estimations** of the estimators are **biased**: The construction of confidence intervals and the hypothesis testing is not reliable.

Detection

- **Graphs**: Identify scatter patterns on x vs. u or x vs. y.



- **Formal tests**: White, Bartlett, Breusch-Pagan, etc. Commonly, H_0: No heteroscedasticity.

Correction

- Use OLS with a variance-covariance matrix estimator robust to heteroscedasticity (HC), for example, the one proposed by White.
- If the variance structure is known, make use of Weighted Least Squares (WLS) or Generalized Least Squares (GLS):
  - Supposing that $\text{Var}(u_i) = \sigma^2 \cdot x_i$, divide the model variables by the square root of x_i and apply OLS.
  - Supposing that $\text{Var}(u_i) = \sigma^2 \cdot x_i^2$, divide the model variables by x_i (the square root of x_i^2) and apply OLS.
- If the variance structure is not known, make use of Feasible Weighted Least Squares (FWLS), which estimates a possible variance, divides the model variables by it, and then apply OLS.
- Make a new model specification, for example, logarithmic transformation (lower variance).

Autocorrelation

The error terms u_t and u_s are correlated across different time periods conditional on the explanatory variables:

$$\text{Corr}(u_t, u_s \mid x_1, \ldots, x_k) \neq 0 \quad \exists t \neq s$$

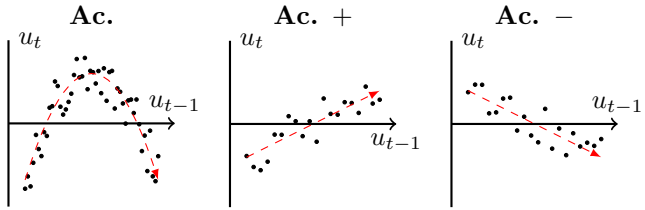
The "natural" context of this phenomenon is time series. This **breaks the sixth (6)** OLS **assumption**.

Consequences

- OLS estimators are still consistent.
- OLS estimators are still unbiased.
- OLS is **no longer efficient**, but still a LUE (Linear Unbiased Estimator).
- **Variance estimations** of the estimators are **biased**: the construction of confidence intervals and the hypothesis testing is not reliable.

Detection

- **Graphs**: Identify scatter patterns on u_{t-1} vs. u_t or make use of a correlogram.



- **Formal tests**: Durbin-Watson, Breusch-Godfrey, etc. Commonly, H_0: No autocorrelation.

Correction

- Use OLS with a variance-covariance matrix estimator robust to heteroscedasticity and autocorrelation (HAC), for example, the one proposed by Newey-West.
- Use Generalized Least Squares. Supposing $y_t = \beta_0 + \beta_1 x_t + u_t$, with $u_t = \rho u_{t-1} + \varepsilon_t$, where $|\rho| < 1$ and ε_t is white noise.
  - If ρ is known, create a quasi-differenced model where ε_t is white noise and estimate it by OLS.
  - If ρ is not known, estimate it by, for example, the Cochrane-Orcutt method, create a quasi-differenced model where ε_t is white noise and estimate it by OLS.